In [64]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.impute import SimpleImputer
```

In [65]:
```
# Load the dataset
df = pd.read_csv("./datasets/uber.csv")

# view dataset
print(df)
```

```
        Unnamed: 0                            key  fare_amount  \
0         24238194    2015-05-07 19:52:06.0000003          7.5
1         27835199    2009-07-17 20:04:56.0000002          7.7
2         44984355   2009-08-24 21:45:00.00000061         12.9
3         25894730    2009-06-26 08:22:21.0000001          5.3
4         17610152  2014-08-28 17:47:00.000000188         16.0
...            ...                            ...          ...
199995    42598914   2012-10-28 10:49:00.00000053          3.0
199996    16382965    2014-03-14 01:09:00.0000008          7.5
199997    27804658   2009-06-29 00:42:00.00000078         30.9
199998    20259894    2015-05-20 14:56:25.0000004         14.5
199999    11951496   2010-05-15 04:08:00.00000076         14.1

                 pickup_datetime  pickup_longitude  pickup_latitude  \
0       2015-05-07 19:52:06 UTC        -73.999817        40.738354
1       2009-07-17 20:04:56 UTC        -73.994355        40.728225
2       2009-08-24 21:45:00 UTC        -74.005043        40.740770
3       2009-06-26 08:22:21 UTC        -73.976124        40.790844
4       2014-08-28 17:47:00 UTC        -73.925023        40.744085
...                         ...               ...              ...
199995  2012-10-28 10:49:00 UTC        -73.987042        40.739367
199996  2014-03-14 01:09:00 UTC        -73.984722        40.736837
199997  2009-06-29 00:42:00 UTC        -73.986017        40.756487
199998  2015-05-20 14:56:25 UTC        -73.997124        40.725452
199999  2010-05-15 04:08:00 UTC        -73.984395        40.720077

        dropoff_longitude  dropoff_latitude  passenger_count
0              -73.999512         40.723217                1
1              -73.994710         40.750325                1
2              -73.962565         40.772647                1
3              -73.965316         40.803349                3
4              -73.973082         40.761247                5
...                   ...               ...              ...
199995         -73.986525         40.740297                1
199996         -74.006672         40.739620                1
199997         -73.858957         40.692588                2
199998         -73.983215         40.695415                1
199999         -73.985508         40.768793                1

[200000 rows x 9 columns]
```

In [66]:
```
df['pickup_datetime'] = pd.to_datetime(df['pickup_datetime'])
# print(df['pickup_datetime'])
df['hour'] = df['pickup_datetime'].dt.hour
# print(df['hour'])
df['day_of_week'] = df['pickup_datetime'].dt.dayofweek
# print(df['day_of_week'])
```

In [67]:
```
# check datasets for more columns we added 'hour' and 'day_of_week' column
print(df)
```

```
        Unnamed: 0                            key  fare_amount  \
0         24238194    2015-05-07 19:52:06.0000003          7.5
1         27835199    2009-07-17 20:04:56.0000002          7.7
2         44984355   2009-08-24 21:45:00.00000061         12.9
3         25894730    2009-06-26 08:22:21.0000001          5.3
4         17610152  2014-08-28 17:47:00.000000188         16.0
...            ...                           ...          ...
199995    42598914   2012-10-28 10:49:00.00000053          3.0
199996    16382965    2014-03-14 01:09:00.0000008          7.5
199997    27804658   2009-06-29 00:42:00.00000078         30.9
199998    20259894    2015-05-20 14:56:25.0000004         14.5
199999    11951496   2010-05-15 04:08:00.00000076         14.1

                pickup_datetime  pickup_longitude  pickup_latitude  \
0      2015-05-07 19:52:06+00:00        -73.999817        40.738354
1      2009-07-17 20:04:56+00:00        -73.994355        40.728225
2      2009-08-24 21:45:00+00:00        -74.005043        40.740770
3      2009-06-26 08:22:21+00:00        -73.976124        40.790844
4      2014-08-28 17:47:00+00:00        -73.925023        40.744085
...                          ...               ...              ...
199995 2012-10-28 10:49:00+00:00        -73.987042        40.739367
199996 2014-03-14 01:09:00+00:00        -73.984722        40.736837
199997 2009-06-29 00:42:00+00:00        -73.986017        40.756487
199998 2015-05-20 14:56:25+00:00        -73.997124        40.725452
199999 2010-05-15 04:08:00+00:00        -73.984395        40.720077

        dropoff_longitude  dropoff_latitude  passenger_count  hour  \
0              -73.999512         40.723217                1    19
1              -73.994710         40.750325                1    20
2              -73.962565         40.772647                1    21
3              -73.965316         40.803349                3     8
4              -73.973082         40.761247                5    17
...                   ...               ...              ...   ...
199995         -73.986525         40.740297                1    10
199996         -74.006672         40.739620                1     1
199997         -73.858957         40.692588                2     0
199998         -73.983215         40.695415                1    14
199999         -73.985508         40.768793                1     4

        day_of_week
0                 3
1                 4
2                 0
3                 4
4                 3
...             ...
199995            6
199996            4
199997            0
199998            2
199999            5

[200000 rows x 11 columns]
```

In [68]:
```
# Drop unnecessary columns
df = df.drop(columns=['Unnamed: 0', 'key', 'pickup_datetime'])
```

In [69]:
```
# check datasets for removal of columns we removed 'first_column with no name', 'key' and 'pickup_datetime' column
print(df)
```

```
        fare_amount  pickup_longitude  pickup_latitude  dropoff_longitude  \
0               7.5        -73.999817        40.738354         -73.999512
1               7.7        -73.994355        40.728225         -73.994710
2              12.9        -74.005043        40.740770         -73.962565
3               5.3        -73.976124        40.790844         -73.965316
4              16.0        -73.925023        40.744085         -73.973082
...             ...               ...              ...                ...
199995          3.0        -73.987042        40.739367         -73.986525
199996          7.5        -73.984722        40.736837         -74.006672
199997         30.9        -73.986017        40.756487         -73.858957
199998         14.5        -73.997124        40.725452         -73.983215
199999         14.1        -73.984395        40.720077         -73.985508

        dropoff_latitude  passenger_count  hour  day_of_week
0              40.723217                1    19            3
1              40.750325                1    20            4
2              40.772647                1    21            0
3              40.803349                3     8            4
4              40.761247                5    17            3
...                  ...              ...   ...          ...
199995         40.740297                1    10            6
199996         40.739620                1     1            4
199997         40.692588                2     0            0
199998         40.695415                1    14            2
199999         40.768793                1     4            5

[200000 rows x 8 columns]
```

In [70]:
```
# Handle missing values
imputer = SimpleImputer(strategy='mean')
df_imputed = pd.DataFrame(imputer.fit_transform(df), columns=df.columns)
```

In [75]:
```
# Split the data into features (X) and target (y)
X = df_imputed.drop(columns=['fare_amount'])  # create new dataset ignoring 'fare_amount' column
y = df_imputed['fare_amount']  # create a series of only 'fare_amount' column
```

In [76]:
```
# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

In [77]:
```
# Standardize the features (scaling)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

In [78]:
```
# Implement Linear Regression
lr_model = LinearRegression()
lr_model.fit(X_train_scaled, y_train)
y_pred_lr = lr_model.predict(X_test_scaled)
```

In [79]:
```
# Implement Ridge Regression
ridge_model = Ridge(alpha=1.0)  # You can experiment with different alpha values
ridge_model.fit(X_train_scaled, y_train)
y_pred_ridge = ridge_model.predict(X_test_scaled)
```

In [80]:
```
# Implement Lasso Regression
lasso_model = Lasso(alpha=0.1)  # You can experiment with different alpha values
lasso_model.fit(X_train_scaled, y_train)
y_pred_lasso = lasso_model.predict(X_test_scaled)
```

In [81]:
```
# Evaluate the models
def evaluate_model(y_true, y_pred, model_name):
    r2 = r2_score(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    print(f"{model_name} - R2 Score: {r2:.4f}, RMSE: {rmse:.2f}")
```

In [82]:
```
evaluate_model(y_test, y_pred_lr, "Linear Regression")
evaluate_model(y_test, y_pred_ridge, "Ridge Regression")
evaluate_model(y_test, y_pred_lasso, "Lasso Regression")
```

```
Linear Regression - R2 Score: 0.0007, RMSE: 10.31
Ridge Regression - R2 Score: 0.0007, RMSE: 10.31
Lasso Regression - R2 Score: 0.0003, RMSE: 10.31
```

In [ ]: